

Hoomie Sweet Home: A Qualitative Study on the Small Multisensory Space "Hoomie" for Inclusion of Autistic Kids in a Primary School

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ABSTRACT

Multisensory environments (MSE) are thought to promote relaxing experiences and learning opportunities for children with neurodevelopmental disorders (NDD), especially autistic kids. However, there seems to be a lack of proposals tailored to school with relaxing goals that address autistic kids' anxiety, while answering support teachers' needs. This paper presents Hoomie, a small relaxing multisensory space. A qualitative study with eleven NDD students and their support teachers was conducted to understand what elements should be taken into consideration to allow as much flexibility as possible to children and teachers, in order to be accessible for the kids with sensory processing dysfunction and adoptable in teachers' routines. Findings suggest that by providing a wide range of possibilities in a few touchpoints, a small multisensory space can keep motivating the children's interaction, while flexibility in activity management, activity parameters, and work method can answer teachers' needs for adoptable tools.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design**; **User studies**; • **Social and professional topics** → **People with disabilities**.

KEYWORDS

interactive multisensory environment, children with special needs, primary school education, well-being, smart space

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1 INTRODUCTION

1.1 Context and related work

Autism Spectrum Disorder is a neurodevelopmental disorder characterized by deficits in social communication and interaction, as well as by the presence of repetitive behavioral patterns and restrictive interests [3]. Up to 95% of autistic children show some degree of sensory processing dysfunction [23], leading to atypical multisensory integration [4] that can impact social functioning [22]. Deficits in sensory integration are coupled with enhanced low-level perception, creating an intense and fragmented world [18]. Accessible spaces should consider how sensory information is displayed to accommodate hypersensitivity and hyposensitivity, and thus allowing a better understanding of the environment [8] and better social opportunities [22].

School is one of the most significant places for socialization with peers, but it is also one of the most chaotic environments for autistic kids [14]. In Italy, primary school kids with a disability attend mainstream classes. They are accompanied by a support teacher who is in charge of promoting inclusion and achieving the goals of the individualized educational plan [2]. While this practice values the social component of the school system, the call for an accessible environment is hard to answer, particularly from a sensory perspective [19]. Moreover, the high occurrence of social and general anxiety in autistic students negatively impacts their well-being [1]. On the other hand, support teachers feel left aside and powerless [10] and up to 47% don't have a specialized formation [19].

Over the years, there has been a constant increase in interactive technologies that address autistic individuals' needs, especially social abilities [15]. Among them, multisensory environments (MSE) focus on the space and the stimuli it displays. MSEs adapt sensory stimuli to the kid's needs, and they're used to facilitate learning through information processing. MSEs not only include the repurpose of a whole room, as in the case of Snoezelen [6], Magika [12], and Mediate [20], but there are also compact versions that intend to transform any room into a multisensory one, like Ahù [13]. Even though spaces based on unstructured play already exist, there seems to be a lack of proposals tailored to school that don't have only educational goals, but also relaxing ones [11].

1.2 Problem statement and research contribution

The lack of sensory accessibility in Italian schools [19] is an obstacle to socialization and education [22], impacting the well-being of

autistic students. Support teachers are often left aside in the school system [10] and lack tools to help their students [19].

The work aims to explore how a small MSE can promote personal and social experiences for primary school kids with a neurodevelopmental disorder (NDD), especially autistic students. Technology enhanced safe-spaces could help make schools more accessible for students that experience sensory overload. By gathering support teachers' perceptions and observing kids' behaviors, we envision the possibility to design a flexible and efficient tool that can improve the teachers' well-being, while addressing the needs of the teachers.

2 METHOD

2.1 Objective and system

The study goal was to understand how kids with NDD, especially autistic kids, interact within a small smart space for relaxation. Moreover, we wanted to explore how teachers perceive an MSE for relaxation in the school setting and what could allow it to become an effective tool.

The research questions were the following:

- How do kids interact within a small MSE? Which factors should be considered to improve its accessibility?
- How do teachers perceive the presence of multisensoriality for relaxation in school? Which factors define the flexibility of a multisensory tool?



Figure 1: (a) Hoomie at school. (b) Relaxation activity. (c) Expression activity. (d) The components of Hoomie.

The investigation focused on Hoomie. Hoomie is a system composed of a tent and an app. On the app, the teacher can select the stimuli to display and create activities. Teachers can plan sessions or they can use it whenever the kid needs a break. To make it as cost-effective as possible for a school, we focused on two main elements: an interactive LED panel and eight tactile panels. The interactive panel is composed of fifty cells with six LEDs and a proximity sensor each. It allows different interaction modalities and a good range of possible activities while being affordable. On each lateral wall, there are four removable tactile panels that help teachers customize their activity by adding another type of sensory information that can be useful for kids that prefer tactile stimuli. We chose materials that can be interacted with different gestures, like grabbing, pressing, and stroking. On the floor, there are pillows and a soft rug that act as additional tactile items while making the space more welcoming and comfortable. Lastly, two colored lights increase the overall illumination.

2.2 Codesign and school study procedures

We took Hoomie to an Italian primary school. The study involved eight classes, eleven kids with an NDD, of which four were autistic, between six and eight years old. Eleven support teachers participated in the sessions, and six of them took part in the codesign activity.

The codesign session aimed to understand teachers' perception of a small MSE in school and their perception of it as a tool. The tent is controlled by the teachers' application on which they can design and plan activities, so we prototyped a paper version of the app creation screens to spark the discussion. Teachers had to imagine an activity and how they would implement it in their routine. They had to start from the sensory profile [5] of an autistic kid, and select and paste on the prototype the visual, auditory, and tactile settings they preferred. Eventually, they shared their results and discussed them.

The main study lasted one week, for a total of five days and five sessions for each kid. Each session was video-recorded to conduct an indirect observation afterward. The observed parameters were related to children's interaction within the space and their reactions, and social interaction between student and teacher. Every session kids could decide whether to enter or not. For three days, kids participated in relaxation sessions based on music listening activities [7]. We aimed to creating a safe environment and to investigate their response to it. They were alone in the tent, to let us observe their unprompted exploration. The speakers played a piece of consonant classical music [7], while the LED panel displayed a soothing animation. Eight tactile modules were present to observe what caught their attention. Every session lasted five minutes. The last two days, kids joined an unstructured expression activity inspired by Mediate [20]. This time, they played together with their teachers by touching the LED panel to play sounds and light the cells. We wanted to see if they created rules and how their teachers responded to them. Every session lasted ten minutes.

3 FINDINGS

3.1 Codesign findings

In the codesign session, teachers imagined on paper a multisensory activity starting from a sensory profile and provided elements. One couple decided the activity goal before creating it, while the others designed a relaxation activity without a predefined objective. They all set the duration to fifteen minutes, as in their experience this is the optimal one. They didn't have problems envisioning an MSE in their routine, even though they never used one, and the school doesn't currently have a similar tool. They imagined its use mostly after morning break and lunch break. These are usually the most challenging moments for the kid because of sensory hyperstimulation and tiredness. The most insidious task they identified is the turn management of Hoomie: Hoomie would be useful for sudden breakdowns, and not only for planned use.

T1: "Everything depends on everything. That's the difficult part. Activities must be created with that in mind: their success depends on the kids' needs, on their mood, on the previous activities, and on what we feel it's right to do in that specific moment."

T2: "A multisensory room is almost impossible to think in this school. I believe all the kids would benefit from it, but they can't

use it together: each one of them has completely different needs and strengths, especially from a sensory point of view. But if we think about multiple small spaces, here and there... That could grant us greater freedom."

3.2 School study findings

As far as the school study, the three relaxation sessions were completed by nine kids out of eleven, as two kids only engaged in two sessions due to illness. In the first session, most of the kids looked occasionally for their teachers, who were outside the tent. Interestingly, the exploration often began at the end of the activity when the music stopped, and the observer and the teacher commented on the experience between them. It seems that initially they didn't know that they were allowed to interact with Hoomie. However, in the following sessions, children progressively felt comfortable enough to interact. Many began to lay down and rest.

While the overall experience was similar for the majority of the kids, two children had a different relationship with Hoomie. One kid with a non-specified NDD was more attracted to tactile panels than others. They didn't just stroke the materials but preferred to tear them off the panel and place them on the floor. Once on the floor, they were able to manipulate them in a more complex way. The kid also used the base of the panel to reflect the light, thus showing a much more exploratory interaction than their schoolmates. The teacher pointed out that their active participation was surprising compared to regular school activities. The other case worth noting concerns an autistic student. Unlike the others, the child was very uncomfortable, especially in the second session: even if sitting inside, they remained with their legs out and for the entire duration of the activity they showed stereotypies, which the teacher specified were not typical. For this reason, in the third session, the teacher was inside the tent, while the observer moved away. Thanks to the presence of the teacher, the kid relaxed, interacted with the space, and lay down to rest.

As for the expression activity, the first session was completed by all the children, while only seven children were able to engage in the second one due to illness. All the children of the first session actively participated. As in the relaxation session, despite the diversity of diagnoses and their severity, the experiences were homogeneous. Children explored the wall with their bare hands and the provided handles. After exploration, they invented and followed their personal game logic. Two children also communicated the rules to their teachers. Some teachers spontaneously followed the students' play and added small educational moments to the game, such as counting, saying the names of colors, or challenging them to make particular movements. Other teachers decided not to join the game and, even if they were inside the space, stood aside. The kid who interacted the least was the most engaged by tactile panels in relaxation activities. The teacher was able to engage them for short periods, but they were not as attracted to the LED panel. Instead, they preferred to play with pillows, which are more satisfying to the touch. Additionally, the kid interacted more with the observer than with their teacher, specifically by hugging them and taking their hand. They ended both sessions halfway through the available time.

4 DISCUSSION

In this exploratory study we observed how children with NDD interact with Hoomie to understand what elements should be considered to allow as much flexibility as possible in a small MSE, since adaptability to children's needs is strictly related to accessibility. Moreover, with a codesign session, we investigated with teachers how small MSEs could become effective tools that support them in their role.

The interaction within a MSE can stem from three different motivations: to stimulate interesting sensations, to change the appearance of the space according to one's preferences, or to explore in search of a goal that makes the activity structured. To make children feel in control, and therefore less worried by uncertainties [14], we need to allow them to decide their own experience [21], and consequently work on how to facilitate the three aspects. The presence of objects with different degrees of interaction and with outcomes with different sensory impacts enable the child's choice. Multimodality is also an important factor [11]. To grant accessibility, designers have to consider the position of objects and their freedom of manipulation. In a small space, it's also crucial to have reachable items without having to stand up. For an interactive screen, activation with different tools and different gestures can better adapt to the child's preferred interaction modality.

To be useful for teachers and for the educational goals they pursue, a technological tool for inclusion must be flexible. First, the system must be flexible to manage, to decrease the perceived effort of use [9]. In addition, it must guarantee the possibility of both planned and unplanned use. Secondly, flexibility must respect the teacher's work method. There seem to be at least two different ways of imagining a multisensory exercise: one tailored to the outcome, and the other tailored to the characteristics of the kid. While in the first case the creation starts with precise intentionality, in the second case the goal is pursued on the spot by capturing the educational cues that naturally arise during the child's experience. Thirdly, the flexibility of multisensory stimuli must be considered. When creating an activity, teachers do not seem to need numerous settings, that indeed make the use of the system more complex, but rather a few customization areas which, however, allow great freedom within them. In this way, the quality of the performance is high, while the effort expectancy is low [9], increasing the chances for adoptability [17].

Flexibility and adoptability shouldn't be achieved at the expenses of the prevention of improper uses that can lead to exclusion, isolation, and even harmful experiences. Since teachers need to be aware of their choices when using technology for inclusion, the system controller, like the app, shouldn't favor carefree use, but instead it should promote conscious decision-making by implementing forcing functions [16]. In addition, the user interface should allow kids to participate as much as possible in the settings selection in order to promote communication and understanding.

5 CONCLUSIONS

Even though Hoomie offers less interaction modalities compared to traditional MSEs, this study suggests that a small MSE can be an answer to both accessibility needs of autistic students and flexibility needs of teachers. Attention should be paid to allow a wide range

of possibilities through a few touchpoints: by doing so, a small MSE can be more affordable and easier to use while it keeps motivating children's interaction. Extensive research should be conducted to understand the efficacy of Hoomie over a long period in improving the students' well-being. It's also important to investigate the position of an MSE inside the school, to understand if and how it influences the experience. Moreover, the impact on the teachers' work routine should also be considered to improve its usability.

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